

Transliterating Filipino Sign Language using Long Short-Term Memory

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I. INTRODUCTION

A. Background of the Study

Communication is vital for social beings, as this is the basis for day-to-day activities. This not only signifies that individuals belong to a particular group but also provides social utilities that ensure the group's survival. Through communication, individuals learn to understand, respect, and empathize with one another, and being deprived of such interaction leads to a breakdown of the human experience. However, according to the Philippine Statistics Authority (2020), in 2020, there are an estimated 1,784,690 individuals who have difficulty hearing. In order for them to communicate effectively, they have learned how to use sign language from an early age. These individuals are compelled to learn FSL from specialized schools or from local deaf communities so they can achieve some semblance of a normal life [1].

Along with the Republic Act 11106, known as "The Filipino Sign Language Act," this has allowed the FSL to be recognized and supported as the official language for all transactions concerning the deaf. Despite this fact, Filipinos are still unfamiliar with FSL, and trying to learn it would require significant training. This has led to a major communication gap for those with disabilities and the majority of people. According to Notarte-Balanquit [2], FSL remains as an understudied sign language in Southeast Asia where she identifies that there are subtle differences in how FSL is taught much like dialects in a spoken language. She notes that the language varies at a national, local, and home level due to: iconic signs, the history of FSL with other languages, and their interaction with other people. This poses a problem for individuals who attempt to learn FSL, as they would need to consider these factors and would often resort to hiring interpreters to bridge that gap.

As such, several studies have been conducted to deal with the communication gap. A number of systems have been developed over the years that employ the use of specialized glove hardware, computer vision, and other hybrid setups that make use of artificial intelligence (AI) models [3][4][5][6]. The hardware-based system makes use of special sensors to obtain relevant information relating to the gestures made by the user while vision-based systems rely solely on the information given by the camera. While hardware-based systems provide 3D datasets that can accurately predict hand gestures, its setup and equipment prove to be costly and the gloves developed may be impractical at times. Most of the recent studies show that, vision-based approaches are more commonly explored since it's cheaper to develop, AI model predictors have been improving and have exploded in popularity, and cameras are more refined nowadays.

The proposed study aims to create a web application that can take in real-time video input of signers then transliterate it for non-signers. The application will make use of a Recurrent Neural Network that is commonly used for ordinal or temporal problems, and is ideal for action recognition. The proposed system will be set up using a web camera that has a clear view of the signer's hand movement and facial expressions in order to annotate certain real-time video inputs. By considering each frame of the video input, the model can predict more complex words in the Filipino Sign Language and can potentially understand an entire phrase or sentence. The scope of the study is to accurately transliterate sentences that are commonly used in conversations.

B. Significance of the Study

Despite having the RA 11106 requiring businesses and government transactions to be able to converse through FSL, the communication gap is still present for signers and non-signers. Addressing this issue, there have been several studies made in previous years, however these studies make use of datasets that are usually static images that are limited to letters of the alphabet or numbers. This study aims to contribute to this growing field by using real-time video inputs as its datasets, which then focus on the user's hand movements and facial expressions. These key features, when extracted, can give valuable insights to further future developments in creating a real-time application that can transliterate a signer's message. This study can also help educate people about the subtle distinctions made by signers that could be missed by inexperienced FSL learners but are fully recognized by computers.

C. Statement of the Problem

Communication gap happens when there is a disconnect in the message being conveyed between two parties. This can cause misunderstandings about the relayed information, or the message itself could be lost in the process. Businesses often put an emphasis on dealing with this issue to avoid miscommunication. However, this can be complicated due to the language barrier for the parties involved, which can lead to reduced productivity. This gap can cause transactional interactions to be inefficient and may cause a waste of resources, such as time and money. On a personal level, miscommunication can lead individuals to hurt their relationships and strain future interactions if it can't be resolved intimately. This becomes difficult when the gap presented is between verbal and nonverbal communication.

A roadmap for the next 20 years of computational linguistics in the Philippines on its research for Filipino Sign Language was presented in a study by Martinez et al. [7]. Since then, numerous systems have been explored for FSL research, however it shows that the Philippines is off-track on the roadmap and FSL remains underutilized when dealing with those who have hearing impairments. The study by Martinez et al. speculated that by 2018, researchers would have a good handle on FSL, a freely available dataset of gestures, and the FSL grammar that is well documented. However, studies since then have only explored simple gestures for alphabet and numeral recognition, and recent studies have tackled nuances of SLR such as non-manual signals and action recognition. Due to the complexity of the field, this study aims to contribute to the research field by creating a platform to transliterate FSL conversational phrases.

D. Scope and Limitation

The study will focus on the computer vision approach, which makes use of the Long Short-term Memory AI model to learn Filipino Sign Language. The study will also make use of quantitative methods to collect and analyze data that is limited to twenty (20) respondents who can use FSL. The study will cover up to 25 sentences that are commonly used in a day-to-day conversation. As the study will focus on its effectiveness for a conversation, it will not consider letters in the alphabet or the first few numbers. This study will not feature the systems that make use of the glove-based approach as well as its hybrid approach since this would require computer engineering expertise and would be costly to recreate.

II. OBJECTIVES

The study aims to determine how effective the Sign Language Recognition platform is in identifying conversational phrases made by the signers.

Specifically, the aims of this study are to:

- 1) Develop an online platform that can take in a real-time video that can translate and display messages of the signer's input;
- 2) Evaluate the accuracy of the application in terms of conveying their message; and
- 3) Determine the user-friendliness of the application by evaluating its ease of use.

III. REVIEW OF RELATED LITERATURE

Several studies about Sign Language Recognition (SLR) have been done since the research made by Martinez and Cabalfin[7] proposing the roadmap for Filipino Sign Language (FSL) recognition. However, some of the studies explored within the past few years limited themselves to image classification of basic gestures and rarely incorporate key features of the signers, such as their facial expressions. Thus, this section presents recent studies of Sign Language Recognition (SLR) and the various methods employed to each application.

A. Non-Manual Signals

Studies over the past years have advanced the field of SLR however, most studies would only concern themselves solely with Gesture Recognition (GR), and rarely would they incorporate non-manual signals [7]. In a study by Rivera and Ong [8], they compiled interviews with FSL professors and showed that much of the context of a sentence lies within the non-manual signals. Insights from their interview revealed that these non-manual signals help convey Affective Facial Expressions (AFE) that tell emotions and feelings and Grammatical Facial Expressions (GFE) that denote the type of sentence as well as degrees of adjectives. These critical information can directly change an individual's understanding of a conversation, even though communication without such details may be possible. By associating facial expressions with several signed words, this will help train the proposed model to be wary of these distinctions.

B. Various SLR Algorithms

1) Convolutional Neural Networks (CNN):

Numerous machine learning algorithms have been in use in SLR for years now, many of which make use of Convolutional Neural Networks. This type of deep learning algorithm is well-suited for visual data, hence its popularity [9]. A study by Cabutaje et al. [3] shows that the FSLR based on CNN achieved an accuracy score of 92% and a validation rate of 93% over the FSL alphabet. Another study made use of Inflated 3D CNN over a dataset of around 240 clips of common FSL words and phrases, which are 20 different gestures in total. After retraining their model layers, their study yielded up to 95% accuracy over their dataset [10]. Though the studies presented for CNN and its deep learning counterpart look promising, ultimately, the model they've employed is determined by the availability of the dataset. In a typical CNN setup, the problem arises with how the model processes information. Its context is bound to a frame, which can create complications for hand movements that are largely temporal. Its datasets are sensitive and are easily interfered with by background noises (such as hats and gloves) and occlusions. Generally, image processing techniques are needed to preprocess datasets to increase accuracy. 3D CNNs address these temporal and spatial issues by processing multiple frames to try and find patterns and associations within a volumetric data. Due to its 3D convolution, the model is computationally expensive, which can cause a bottleneck in the application and oftentimes limit datasets to shorter videos. The model becomes less ideal as inputs become longer and dependencies built between frames become less associated with each frame.

2) Recurrent Neural Networks (RNN):

Considering the sequential nature of the datasets, a deep learning model should be considered to perform action recognition. Recurrent Neural Networks (RNN) are specifically designed to handle sequential data, find dependencies, and interrelate complex information to create a sequential output (Amazon Web Services, n.d.). A study by Tupal et al. [11] shows that the use of RNN-based models in a SLR application has proven to be successful, with a Gated Recurrent Units

(GRU) model achieving an 86.74% accuracy rate over 20 FSL signs. Their study also featured a popular RNN model, Long Short Term Memory (LSTM), that performed behind the GRU model with an accuracy rating of 63.86%. While the GRU model outperformed the LSTM model over the given dataset, there may be factors to be considered that could skew their results. LSTMs are a special kind of RNN model designed to limit the flow of information through gates that allow for long-range dependencies, while GRUs make use of simpler gates and can be referred to as simpler LSTMs [12]. Though the performance of the GRU model seems favorable, it may run into a problem where the sequence of data is so long that it fails to identify dependencies. In the study by Tupal et al., their research dataset accommodated short phrases that may not have needed too much action recognition, which favored the use of GRUs. The use of LSTMs, however, poses a different set of problems. In a study by Mittal et al. [13], they gave great importance to the modification of LSTMs to have an additional reset gate in order to control the flow of contextual information between signs. Since the training of the model will make use of signed sentences, long sequences with barely any context to each individual sign will prove fatal and will make the model ineffective.

The preceding reviews show the different implementations of SLR and the problems faced by each setup. Addressing these issues has become an arduous task that spawned the “continuous development of deep learning techniques” that has allowed the advancement of the SLR field [9]. The applications discussed above show promising results in transliterating their respective datasets, however creating a robust SLR application should be able to convey the signer’s message and emotions in order for us to fully understand them.

IV. METHODOLOGY

The setup and tools that will be used in this research are described in the following section. The libraries needed to train and build the model are all in Python. Most of the processes needed for the research will be shown in key diagrams that explain the overview of certain steps in a given process. This section will also describe the web integration of the model and its usage.

A. Preliminaries

Most of the libraries needed for training and building the model will all be in Python, where it can make use of Tensorflow and Keras library functions. The Long Short-Term Memory model described in the previous sections will be built using Tensorflow Keras, which is an open-source library that allows for the development of learning machines and deep learning algorithms. To handle real-time computer vision tasks, OpenCV will be used to help analyze and manipulate frames of data. Data acquisition and collection, along with the building and training of the model, will be further explained in the following sections through workflows.

1) Workflow for Building and Training the Model:

The proposed application for Filipino Sign Language Recognition will be described in this section. The implementation and libraries used will be detailed here along with the steps on how the model is built. Figure 1 shows the generalized steps in building the model.

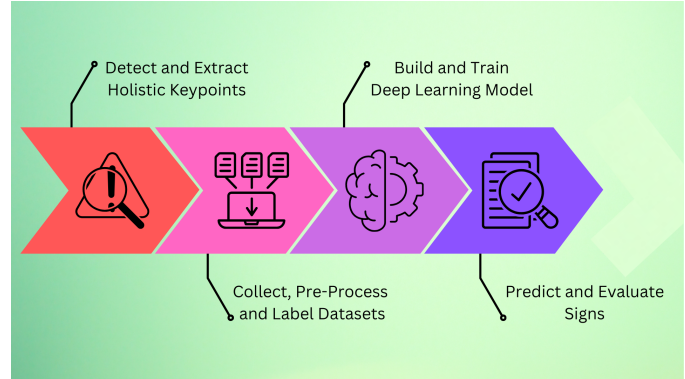


Fig. 1: Activity Diagram / Flow

Data Acquisition

The dataset will be created from online Filipino sign language videos that teach basic words and phrases. These videos will be fed into the MediaPipe Holistic model. The model is a popular pipeline that optimizes the simultaneous detection of the hands, face, and posture of the body as well as tracking of their landmarks (Google AI Developers, n.d.). Figure 2 shows the creation of the landmarks of the subject in their pose, along with their face and hands, which will be tracked on each frame of the video.

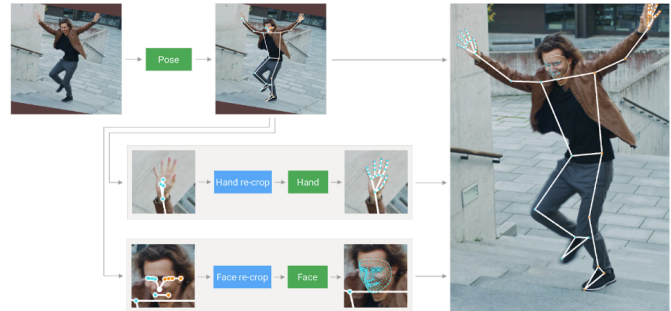


Fig. 2: MediaPipe Holistic Sample Image

Pre-Processing and Data Collection

The landmarks created from the MediaPipe Holistic model will enter pre-processing where key frames of an action will be extracted, compiled, and labeled. These collected information will be categorized into their action label, each action corresponds to a word in the FSL. Each video will have around 30 frames to work with and ensure uniformity. The dataset for each action will be split into a 70-30 split for training and testing, respectively.

Building and Training

The Long short-term model will be built from Tensorflow Keras and will be trained and tested using the landmarks of

the MediaPipe Holistic model. The dataset will be split into 70% for training and 30% for testing the proposed model. The development of the model will be monitored until it reaches an acceptable level of accuracy. To monitor the progress of training, TensorFlow allows users to check the relative progress using TensorBoard. Once the training is done, it can then move on to testing the model to check whether it requires further training or not. Figure 3 depicts the process of training and building the proposed model.

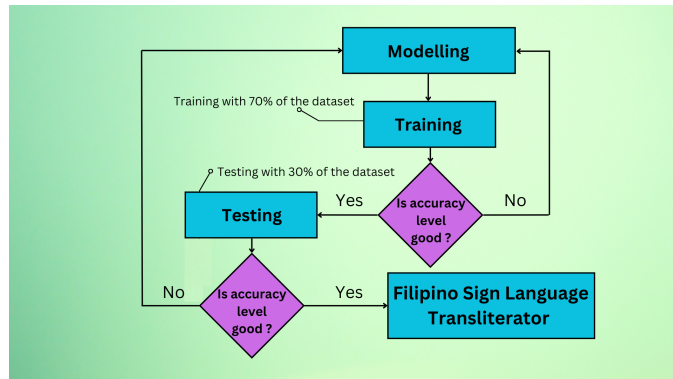


Fig. 3: Filipino Sign Language Transliterator Model Flowchart

2) Web Integration:

The model will be accessible as a web application. For the framework, Flask will be used to easily build the application as well as handle API calls with minimal overhead. As a proof of concept for its ease of use, there will be minimal styling involved. This proposed application will be deployed using Heroku, a free hosting platform.

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